

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

DATA SCIENCE LAB FOR SMART CITIES

FINAL ESSAY

Where Crime Meets Data: Spatio-Temporal Patterns, Neighborhood Disadvantage, and the Fairness of Predictive Policing in Chicago

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September 14, 2026



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Abstract

Urban crime is unevenly distributed across cities and is often concentrated in neighborhoods shaped by socioeconomic disadvantage. This essay examines crime and public safety in Chicago through four questions: where and when crime concentrates, how socioeconomic conditions relate to crime, whether crime can be predicted, and whether such predictions are fair across neighborhoods. Using 2020–2024 Chicago crime records and American Community Survey data for the city's 77 community areas, the study identifies significant crime hotspots on the West and South Sides. Socioeconomic conditions are significantly associated with local crime rates, although multicollinearity limits the interpretation of their independent effects. Predictive modeling provides only modest gains over a historical baseline, with recent crime patterns accounting for most predictive performance. The fairness audit finds no significant evidence of greater prediction errors in disadvantaged neighborhoods, although this finding is limited to neighborhood-level predictions of reported crime and does not address broader risks of predictive policing. Overall, the findings suggest that data-driven crime analysis can provide useful information for urban policy, but statistical fairness alone is not enough to justify its use in policing. Transparency, human oversight, and independent auditing remain necessary to ensure that predictive tools do not reinforce existing inequalities.

Part 1 — Problem, Literature, Indicators

1. Introduction

Crime and public safety are one of the most important challenges that smart cities face at today's age. It is a foundational priority for the stability of society. It affects everyday life of every citizen, their physical safety, quality of life, mobility and the perception of their neighborhoods. Urban crime is not evenly distributed across an area, but it is concentrated in particular locations depending on time, crime type and characteristics of the neighborhood. For example, a research paper on robbery in Chicago found that crime is spatially concentrated, with robbery clusters occurring both across entire neighborhoods and around specific micro-places. [1] The increasing availability of urban data has also created opportunities to use tools to allocate police resources. These models trained on historical crime data can help identify areas where crime is concentrated and have better efficiency at allocating police resources. However, it may also reinforce existing patterns of inequality.

This essay will analyze the city of Chicago because its extensive open-data infrastructure provides detailed information on crime location, timing, crime type, and arrests, while publicly available socioeconomic data allow these patterns to be examined in relation to neighborhood conditions. Chicago is one of the largest cities in the United States with a population of approximately 2.73 million people separated in 77 Community areas with a variation of socioeconomic conditions and provides a suitable geographic structure. [2] It will also consider whether the predictive tool harms or helps the city.

The project is guided by four research questions:

1. Where and when does crime concentrate in Chicago, do these patterns form statistically significant hotspots/coldspots or are they random?
2. What is the correlation between the socioeconomic conditions and the local crime rates?
3. Can historical spatio-temporal and socioeconomic features predict crime rates across Chicago neighborhoods?
4. What are the fairness and policy implications of this model — does it over- or under-predict risk in disadvantaged neighborhoods, and how should that shape its use in policing decisions?

In order to answer these concerns, the essay first examines the empirical studies and criminological theories that link crime to neighborhood characteristics. Next, it looks at the literature on algorithmic fairness and predictive policing. The indicators used to assess socioeconomic conditions and crime trends are then described, along with their limitations. The second section of the essay uses predictive modeling, socioeconomic correlation analysis, spatiotemporal analysis, and a fairness evaluation to apply these ideas to Chicago data. By doing this, the study makes a connection between the use of urban data to analyze crime and the broader issue of how smart city technology may be applied without worsening already existing social disparities.

2. Social Significance

Crime affects not only the victims but is a social and economic challenge that includes all the resident's wellbeing, trust in institutions, property values, and everyday mobility. Research has shown that crime is concentrated in socially and economically disadvantaged neighborhoods and is increased even during periods of overall crime decline. [3]

Furthermore, high levels of crime can make residents feel unsafe and discourage people from living, investing, or doing business in affected neighborhoods. When these patterns persist, they can contribute to lower housing demand and reduced investment, potentially reinforcing socioeconomic differences between neighborhoods.

3. Literature Review

3.1 Classic criminology

Social Disorganization Theory explains the differences in crime between neighborhoods and it was developed by Chicago School researchers Shaw and McKay in their work, *Juvenile Delinquency and Urban Areas*. [4] It was observed that delinquency was not randomly distributed across Chicago. Instead, it was concentrated in particular neighborhoods, especially areas characterized by low economic status, residential instability, and population turnover. [5] These patterns remained relatively stable even as the racial and ethnic composition of neighborhoods changed, suggesting that neighborhood conditions themselves played an important role in shaping crime patterns. [5]

This theory was later expanded through the concept of collective efficacy, defined as social cohesion among neighbors combined with their willingness to intervene on behalf of the common good. It is linked to reduced violence. [6] Sampson, Raudenbush, and Earls used survey data from 8,782 residents across 343 Chicago neighborhoods and found out that collective efficacy was negatively associated with variations in violence. [6] In the meantime, the relationship between concentrated disadvantage, residential instability, and violence was largely mediated by collective efficacy. Ultimately rather than assuming that socioeconomic disadvantage directly causes crime, in this essay it is analyzed whether characteristics such as poverty, income, unemployment, and education are associated with differences in crime rates across Chicago neighborhoods.

Another theory by Wilson and Kelling called Broken Windows, proposed that visible signs of disorder, such as broken windows, vandalism, graffiti, and abandoned property, can signal that an area lacks social control and may contribute to further disorder and crime. [7]

3.2 Contemporary Socioeconomic Correlates of Crime

Recent research done by Yuri Chung in 2026 through a multiple linear regression has shown that poverty has the strongest effect of violent crimes in the USA, with unemployment in second place. [8]

Additionally similar results were noticed in a study of Houston neighborhoods, identifying unemployment, poverty, and academic achievement as important predictors. Importantly it found that the correlations between race and involvement in crime were largely a result of underlying socioeconomic factors rather than race itself. [9] This is consistent with the present study's focus on socioeconomic conditions rather than demographic characteristics as explanatory variables.

In a study of neighborhoods in 2026, only the **composite Hardship Index** was a significant socioeconomic predictor of crime, whereas poverty, unemployment, and education were not significant on their own. [10] This suggests that a disadvantage may be mediated by the cumulative effect of a number of socioeconomic conditions rather than one factor alone. It implies that the relationship between socioeconomic disadvantages and crime can differ over time. [3] Hence, researchers have often measured concentrated disadvantage by combining several indicators including poverty, unemployment, public assistance, family structure, age, and residential stability. [11]

3.3 Predictive Policing and Algorithmic Fairness

In recent years more research is being done to analyze if the predictive tools built on crime data can harm or help the neighborhoods. In particular, Ziosi and Pruss did a case study for Chicago and found out that police, researchers and the community can interpret the same model results differently. [12]

Taking a more analytical stance, a 2023 study in Synthese uses Chicago's predictive policing tool as its central case and argues that observed bias stems less from the algorithm itself than from the institutional and human decisions embedded in the data it was trained on. [13]

Lum and Isaac (2016) highlight the main risk: a feedback loop. [14] If a neighborhood is heavily policed, it produces more recorded arrests. The model may then interpret these arrests as evidence of higher crime, leading to even more policing in the same area. Brantingham and Mohler provide a more nuanced view, finding that predictive policing can produce biased arrest patterns, but that the effects are not always as strong as critics suggest. [15]

3.4 Gaps and Controversies in the Existing Research

Several gaps in the literature directly motivate this essay. First, findings on socioeconomic predictors are inconsistent, even in Chicago. Poverty and unemployment are significant in some studies but not in others when a composite Hardship Index is used. This suggests that how disadvantage is measured can affect the results, which is why this essay examines both individual socioeconomic indicators and a combined measure.

Furthermore, research on predictive policing and algorithmic fairness rarely combines socioeconomic disadvantage with a direct quantitative fairness analysis. Much of the existing work is qualitative, case-based, or focused on proprietary systems, making it difficult to independently reproduce the results. This essay addresses these gaps by connecting socio-economic and spatial analysis with a transparent predictive model and fairness audit using public Chicago data. It therefore examines not only where crime is concentrated and which factors are associated with it, but also whether the resulting predictions and errors disproportionately affect disadvantaged neighborhoods.

3.5 Ethical and Social Implications of the Problem and Indicators

All indicators used in this study are derived from officially recorded crime, not all actual crimes that occur. It's important to take into account that not every crime is reported to the police. There are many reasons such as: distrust of the police, previous negative experiences, or the belief that reporting will not lead to effective action. Research shows that people in disadvantaged neighborhoods may be less likely to report crimes. [26] As a result the difference in crime between disadvantaged and wealthier neighborhoods could be larger than the official data suggests and recorded crime should not be treated as a perfect measure of actual crime. The official crime data shouldn't be ignored, since it is the most consistent citywide source available. Instead, the results of this study should be understood as measures of recorded crime and enforcement activity, rather than perfect measures of actual crime or lived safety.

A further limitation is the level at which the data are measured. Both the crime and socioeconomic indicators are aggregated at the community-area level. The findings show that neighborhoods with higher poverty or unemployment tend to have higher recorded crime rates, but they do not show that people who are poor or unemployed are more likely to commit crime. Making this connection would be an example of the ecological fallacy, where patterns observed across areas are incorrectly applied to individuals. This distinction is important when considering the policy implications of the study. **The findings can support decisions about investment and resources at the neighborhood level, but they should not be used to make assumptions or take action against individuals based on the characteristics of the area where they live.**

4. Indicators

4.1 Crime rate per capita

Crime rate per capita measures the number of reported crimes over the population of a neighborhood. Using the rate instead of only the crime counts makes different sizes of neighborhoods more comparable. This study will use it as the main measure of neighborhood crime as it is commonly used in criminological research.

4.2 Crime density

Crime density or hotspots will examine where crimes are concentrated in the city. Also, it will be used to analyze how these hotspots are scattered either together or randomly. This essay will use spatial methods to identify the crime density. As well as Local Moran's I, the standard tools for testing this at the local level (each individual area) This indicator is important to answer RQ1 because it examines spatial and temporal distribution of crime.

4.3 Socioeconomic indicators

Socioeconomic indicators measure structural disadvantages using variables: poverty rate, median income, unemployment, and education. In this study, these indicators will be used to analyze whether the socioeconomic conditions are associated with crime in different neighborhoods. A possible limitation is that the dataset ACS where a 5-year estimate is gathered may not capture rapid changes in neighborhood conditions.

Part 2: Data Analytics and Policy

6. Data sources and wrangling

This study uses three main data sources. First the crime data is taken from the City of Chicago's Crimes(2001-present) dataset through the Socrata Open Data API. [16] [17] The dataset contains 1,186,999 crime incidents, which is consistent with Chicago's typical annual number of recorded crimes. It was taken only from January 2020 till December 2024 to match the other dataset ACS 5-year estimates, so the crime and socioeconomic data refer to the same period. It is worth mentioning that the period **includes COVID-19 years rather than being treated as anomalies**, it is discussed as part of the results.

Secondly, as it was mentioned beforehand ACS (American Community Survey) dataset is the socioeconomic data used in the study. [18] It uses a 5-year estimate, and the latest data was taken from 2020 till 2024. The Census API that the dataset was called, returned data for all of Cook County and 1332 census tracts were initially collected. [19] Using data from our third source Tract Boundaries geometries from the Census Bureau's TIGERweb system, a spatial test was done with representative

points relating to the 77-community area of Chicago using a "within" predicate. [20] It resulted in a clean many-to-one join (782 tracts matched to exactly one community area each, since community areas were originally defined as whole aggregations of tracts) and correctly excluded 550 tracts falling outside city limits as suburban Cook County.

The tract-level data were aggregated to the community area level. Population, poverty, labor force, and education counts were summed, while median household income was calculated as a population-weighted mean because medians cannot be summed. Crime data were cleaned by checking dates and community area codes and converting the arrest variable to a Boolean value. They were then aggregated by community area and month, producing total crime, arrest counts, and crime counts by type. The resulting monthly crime data was joined with the ACS data, so the same socioeconomic values are repeated across the 60 months for each Community Area. One group represented crimes with missing or unassigned community area information and was removed. The final dataset therefore contains 4,620 observations across Chicago's 77 Community Areas. Some analyses use the original crime records because monthly aggregation removes certain details such as timestamps.

7. Exploratory spatio-temporal analysis and visualization

This section answers RQ1 by examining where and when crime concentrates across Chicago, and testing whether any observed spatial concentration(hotspots/coldspots) reflect a statistically genuine pattern rather than random variation.

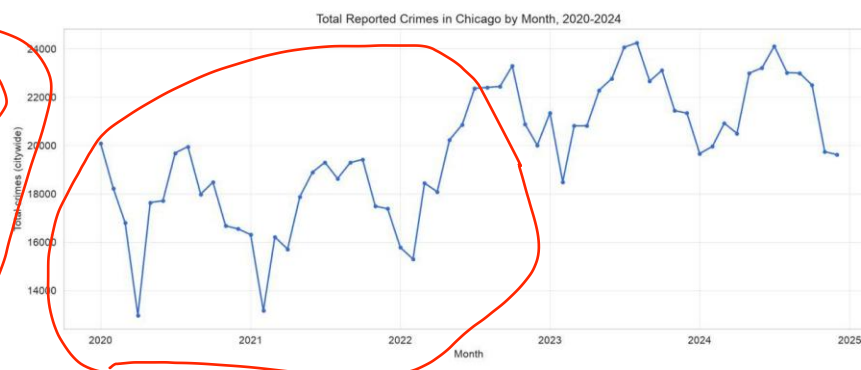


Figure 1 Total Reported Crimes by month

In figure 1 the citywide monthly time series is shown. It is noticed a consistent seasonal cycle with a winter trough, rising in spring and peaking in summer. This is consistent with routine activity theory that in warmer months it increases opportunities for crime. [21] It is striking the drop in spring 2020 which is the deepest point in all five years corresponding to COVID-19 lockdowns. The research shows that the

pandemic restrictions suppressed crime types that required contact or public presence. From 2021 onward, crime rises gradually before stabilizing in 2023–2024. [22]

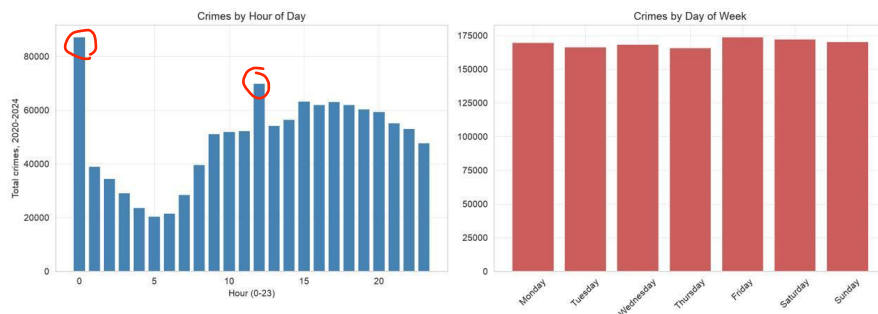


Figure 2 Crimes by hour-of-day and day-of-week

In figure 2 crime is flatly distributed in all days of the week but more noticeably an unusually large spike at midnight was illustrated at the hour-of-day distribution. The midnight spike is not a genuine behavioral pattern as the police recording system defaults to a 00:00 timestamp when the crime's exact time is unknown or reported after the fact. [23] It is demonstrated that at noon is the second peak and at 5 o'clock the lowest crime rate.

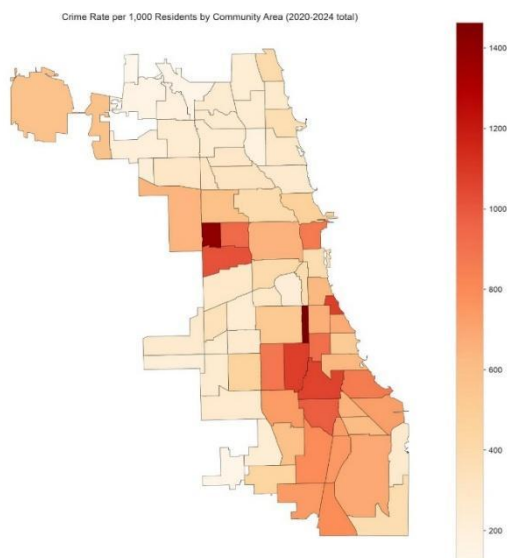


Figure 3 Total crime/1000 residents per community area

In figure 3 a choropleth of raw crime rate per capita by community area shows increased rates across the West Side and South Side and lower rates on the North Side and outer Southwest. This descriptive view is useful but provisional because the crime per capita rates in small population areas can be statistically unstable.

Global Moran's I statistic was computed to test if elevated crime rates cluster spatially rather than randomly and confirmed positive spatial autocorrelation. Then Local Moran's I identifies 16 statistically significant hotspots and 16 coldspots as seen in figure 4. Notably, the two community areas with the single highest raw crime rates, Fuller Park and Oakland, do not appear as significant hotspots, while several areas with lower raw rates do. This reflects the small area problem in spatial statistics. [24] For example, Fuller Park's population is small enough that its per-capita rate is highly sensitive to changes in raw crime counts, producing an unstable estimate rather than a genuine spatial pattern. The same logic explains O'Hare's classification as a "High-Low" outlier, given its minimal residential population. Once filtered,

the result is two large, spatially contiguous hotspot clusters: a West Side cluster (East/West Garfield Park, North Lawndale) and a larger South Side cluster (Washington Park, Woodlawn, Englewood, Grand Boulevard, Kenwood, Greater Grand Crossing, Chatham, Avalon Park, Calumet Heights, South Shore, Auburn Gresham, Roseland, Burnside). The coldspots were concentrated on the North Side and outer Southwest. This spatial pattern closely mirrors Chicago's documented history of redlining and disinvestment, providing an empirical link back to the social disorganization literature reviewed in Part 1. [25]

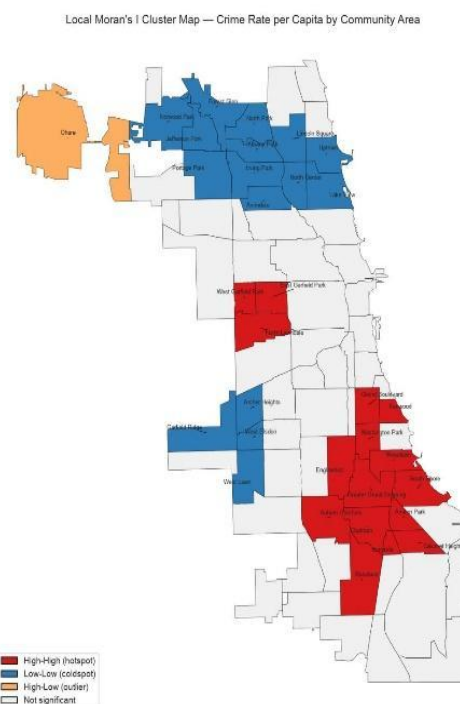


Figure 4 Cluster Map

8. Correlation analysis: Socioeconomic Indicators and Crime

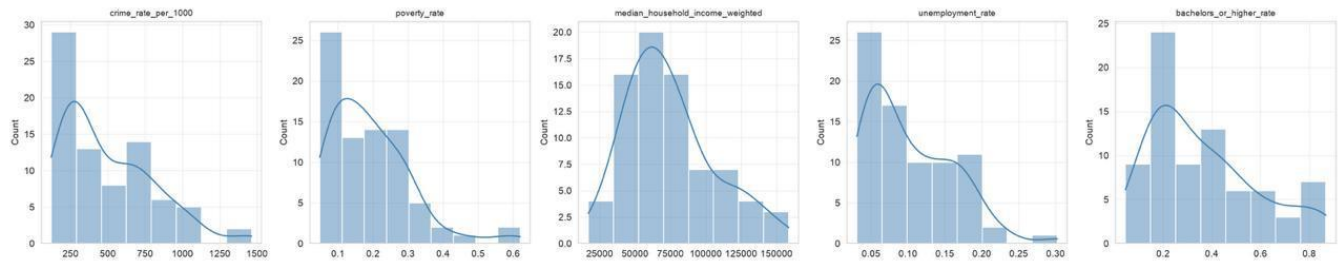


Figure 5 Socioeconomic indicators

To test RQ2, the five-year crime rate per capita for each community area was compared with the four socioeconomic indicators from Part 1 (Figures 5–6). Crime was positively related to poverty ($r = 0.778$) and unemployment ($r = 0.809$), and negatively related to median household income ($r = -0.649$) and bachelor's-degree-or-higher attainment ($r = -0.358$), with all p -values below 0.002. Because the variables are right-skewed, both Pearson and Spearman were calculated. It produced very similar results, suggesting that the findings are not driven by a few extreme neighborhoods.

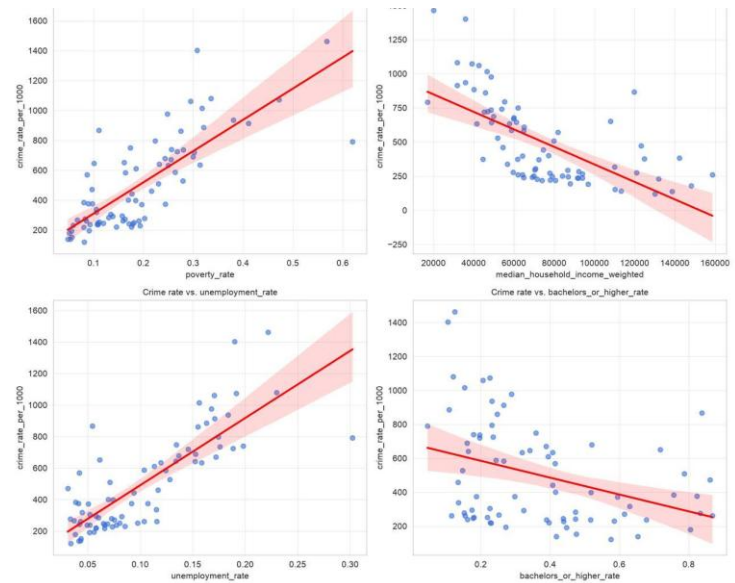


Figure 6 Socioeconomic indicators

The difference between the 16 significant crime hotspots and 16 coldspots provides a clearer interpretation of these relationships. Hotspots had an average poverty rate of 27.8%, compared with 10.5% in coldspots. Their median household income was approximately \$47,000, compared with \$94,500 in coldspots, while unemployment averaged 16.5% compared with 5.6%. All differences were statistically significant at $p < 0.0001$. These results provide evidence that areas with greater socioeconomic disadvantage tend to experience higher crime levels.

A multiple regression using all four indicators explained 72.3% of the variation in crime rates ($R^2 = 0.723$). However, the individual coefficients should be interpreted cautiously because the indicators are strongly correlated with each other, as shown by the elevated VIF (Variance inflation factors) values (3.39–6.73). Only unemployment and education remained statistically significant, while poverty and income lost significance. The education coefficient also changed from negative to positive, which is likely a multicollinearity effect rather than evidence that higher education increases crime. Overall, the results show that socioeconomic disadvantage is strongly associated with crime, but the individual indicators overlap too much to be separated reliably. This supports treating disadvantage as a combined socioeconomic condition, which motivates the use of a composite measure in the predictive model developed in the next section and proving the existing research done in part 1. [10]

9. Predictive modeling of crime risk

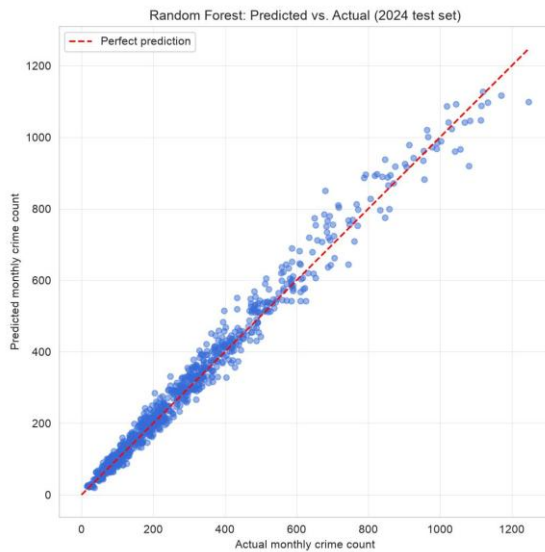


Figure 7 Random Forest: Predicted vs Actual

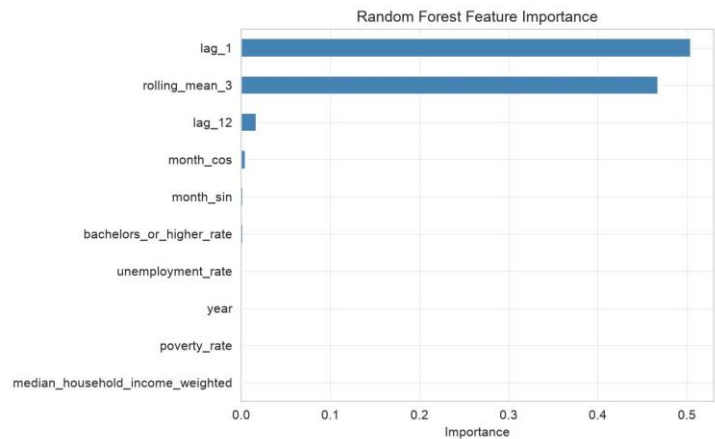


Figure 8 Random Forest Feature Importance

RQ3 was addressed in this section by training a Random Forest regression using the community area panel. The goal was to predict crime counts from previous crime levels and socioeconomic indicators. A simple baseline using only each area's recent crime history performed very well ($R^2 = 0.969$; MAE = 28.93). The Random Forest improved performance modestly, reaching $R^2 = 0.981$ and MAE = 22.08, a 23.7% reduction in error. This suggests that monthly crime levels are highly persistent and that socioeconomic variables provide only a limited improvement once recent crime history is known.

Feature importance helps explain this result. The previous month's crime count and the three-month rolling average together account for more than 95% of the model's predictive importance, while the four socioeconomic indicators combined account for less than 1%. This does not contradict the results from RQ2. The correlation analysis shows that socioeconomic conditions are associated with different crime levels across neighborhoods, while the predictive model shows that recent crime history is much more useful for predicting the next month's crime count.

Also, a robustness check was done to test if the 2021, pandemic year, was distorting the results.

	MAE(all months)	R^2 (all months)	Mean residual,top 3% months	MAE, top 3% months
Model including 2021	22.08	0.981	-18.31	44.09
Model excluding 2021	23.24	0.979	-8.87	47.18

It produced similar results except for the top 3% of observations mean residual revealed a sharper distinction. Both versions systematically under-predict this extreme tail, but the model including 2021 does so, nearly twice as severely. Excluding 2021 reduces this bias but does not eliminate it, and slightly raises absolute error at the tail, since it also removes training data. The deeper cause is structural because only 2% of training rows in either version reach this high-crime range, leaving the model too few extreme examples to anchor an equally extreme forecast to, so it pulls predictions toward mid-range values. The main limitation is therefore not average predictive accuracy but performance in extreme cases.

The model is generally accurate for typical months but tends to underestimate the highest-crime periods. This raises an important fairness question: whether these prediction errors are also concentrated in particular disadvantaged neighborhoods. The next section addresses this question through the fairness audit.

10. Fairness audit of the prediction model used

To answer research question 4 regarding whether the model developed, over- or under- predicts crime in poorer neighborhoods and what that means for potential use in policing decisions, was tested in four different ways. This makes it less likely that a potential fairness problem would be missed by using only one test.

Firstly, each socioeconomic indicator was tested separately against the model's prediction error. None showed a significant relationship. For example, the correlation between poverty rate and prediction error was very small and non-significant ($r = 0.053$, $p = 0.645$). The same result was found using both Pearson and Spearman correlations, suggesting that the result was not caused by unusual values or outliers. The four socioeconomic indicators were then tested together in a regression model. Again, they explained very little of the variation in prediction error ($R^2 = 0.045$, $p = 0.505$).

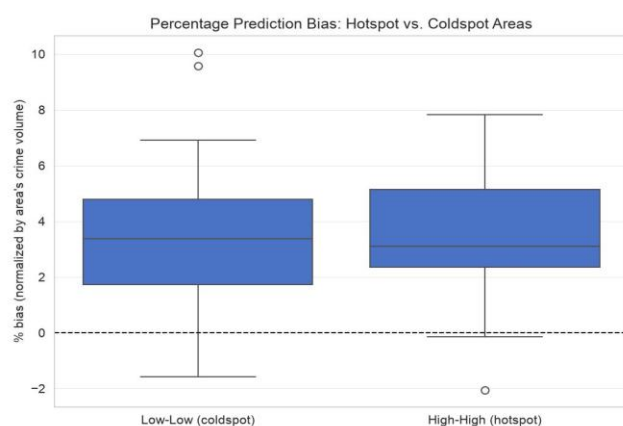


Figure 9 Percentage Prediction Bias

Next, prediction errors were compared between the hotspot and coldspot areas identified in Section 7. As shown in figure 9, the two groups had very similar median bias (approximately 3.1% for hotspots and 3.4% for coldspots). The difference was not statistically significant ($p = 0.73\text{--}0.77$). Finally, the regression was repeated separately for hotspots and coldspots to check whether socioeconomic differences mattered only in high- or low-risk areas. No significant relationship was found in either group.

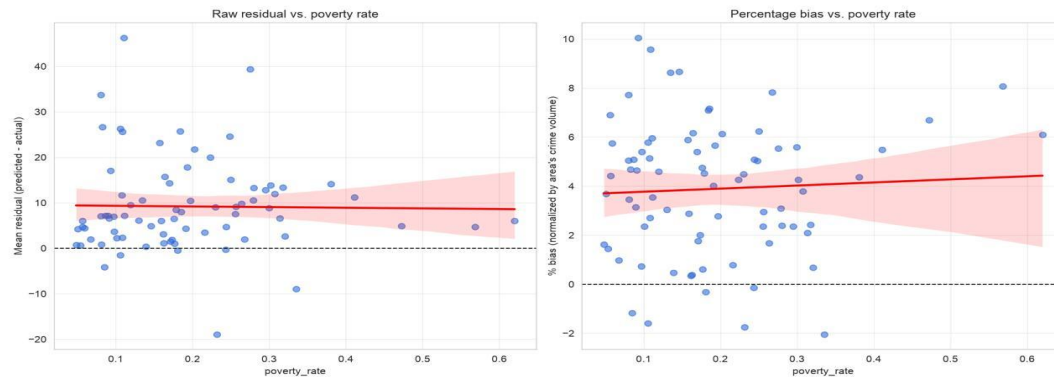


Figure 10 Raw residual vs poverty rate/ Percentage bias vs poverty rate

One finding initially appeared concerning. The model under-predicted crime during the months with the highest crime counts. However, further analysis showed that this was mainly a scale effect rather than evidence of socioeconomic bias. Larger community areas naturally have larger crime counts, so their prediction errors can also be larger in absolute terms. Figure 10 shows this clearly. The left panel presents the raw prediction errors against poverty rate. The relationship is flat and non-significant, but the errors vary widely because of differences in area size and crime volume. The right panel uses prediction error as a percentage of each area's crime volume. After this normalization, the scale effect is reduced and the relationship remains flat and non-significant. The apparent under-prediction was only about -1.5% , and the areas contributing most to the error included both disadvantaged areas, such as Austin, and more affluent areas, such as Near North Side and Near West Side. This suggests that the larger errors were related more to crime volume and area size than to neighborhood disadvantage.

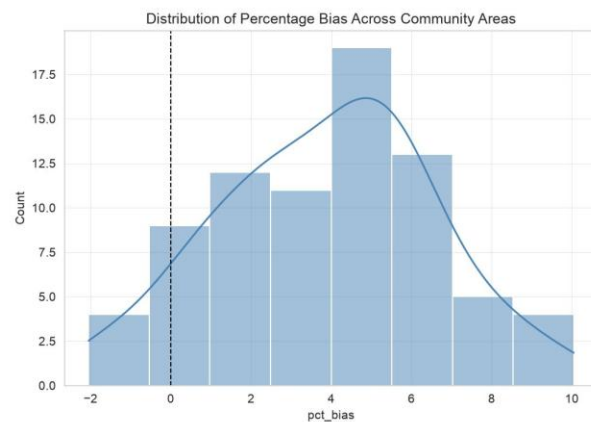


Figure 11 Distribution of Percentage bias across community areas

Figure 11 shows the distribution of normalized prediction bias across all 77 community areas. The distribution is centered around 4–5%, with a slight right skew, but there is no clear separation between disadvantaged and affluent areas.

Overall, the tests provide no statistically significant evidence that the model's prediction errors are systematically larger in disadvantaged neighborhoods. This is an important result, but it needs to be interpreted carefully. The audit measures one specific type of fairness, whether prediction errors are distributed similarly across neighborhoods with different socioeconomic characteristics. It does not test other forms of algorithmic fairness, such as equalized odds or demographic parity, and it does not determine whether the model would be used fairly in practice.

11. Policy recommendations

Target investment at the cluster level, not the neighbourhood level. As analyzed in RQ1 using the Local Moran's I, we found two large hotspot blocks (West Side: Garfield Park areas and North Lawndale; South Side: 13 connected areas) instead of isolated spots. This leads us to a conclusion that when socio-economic disadvantages and crime cluster together instead of staying as isolated blocks, intervention on single neighbourhoods risks just displacing the problem to untouched areas near it instead of actually solving the problem.

Resources should not be allocated based on raw crime-rate rankings alone. Raw crime rates should not independently determine resource allocation because areas with small population can produce misleading high rates from relatively few incidents. In our analysis we noticed that Fuller Park for example, ranked first by crime rate despite having a population of only about 2270 people and not being part of a statistically significant hotspot. Meanwhile a larger area such as Englewood showed steady and statistically meaningful concentrations of crime. A statistically validated hotspot detection like we used in our analysis would be a cheap way to prevent potentially large misallocations of limited budgets.

Neighbourhood disadvantages should be addressed as composite conditions and not through single-variable intervention. Policies should address poverty, income, unemployment and related socio-economic factors together instead of treating them as separate problems. In our analysis, the high VIF values of 5.3-6.7 show that these factors are strongly connected to each-other and represent different aspects of the same broader condition. Addressing only one factor independently may have a limited effect compared to addressing them as a group of factors. Addressing these factors together, other than giving better overall results, avoids the need to label one single factor as the most responsible for crime.

If a predictive tool is used operationally, it should have adequate human oversight and not automated cutoffs. Automated predictions should only inform decisions but the final decision should go through human reviewing. Although in our analysis the model achieved an R^2 of 0.981 the seasonal baseline was

already at 0.969, meaning that most of the accuracy comes from already existing crime patterns. The model's reduction in MAE of 23.7% is meaningful but the average error of about 22 crimes per month may still affect resource decisions. One more reason for adequate human oversight is the tendency of the model to under-predict the high-crime periods, leading to a risk of allocating less resources than needed to these areas.

If a predictive tool is set to be used it should require independent fairness audits before deployment. Predictive policing systems should pass independent fairness audits before deployment and should use reported crime data instead of arrest data as a training target. Chicago's SSL operated for about seven years before it was shut down by an independent audit, showing the risks of identifying problems after years of use.[27] Also predictive systems shouldn't lean on arrest data as a target as it may cause a feedback loop, causing areas with historically higher police presence to show higher accuracy and reinforce existing patterns of policing. Using reported crime counts avoids this problem and with higher independent oversight we can improve accountability and trust in communities that have already experienced the effects of predictive policing systems.

12. Conclusion

This essay examined the spatial distribution of crime in Chicago, its relationship with socioeconomic conditions, its predictability, and the fairness of the resulting predictive model. The main findings are:

- **Crime is spatially concentrated.** Global and Local Moran's I identified two statistically significant hotspot clusters on the West and South Sides. These provided more reliable evidence of concentrated crime than raw crime rates alone, which can be misleading in areas with small populations.
- **Socioeconomic disadvantage is strongly associated with crime.** Poverty, unemployment, income, and education were all significantly related to local crime rates, with clear differences between hotspot and coldspot areas. However, the strong correlation between these indicators makes it difficult to isolate the effect of any single factor.
- **Prediction relies mainly on recent crime history.** The Random Forest model provided only a modest improvement over a simple historical baseline. Recent crime patterns contributed most to its predictive performance, while socioeconomic variables added relatively little. The model's tendency to under-predict high-crime periods was also found to be related primarily to crime volume and area size rather than neighborhood disadvantage.
- **No significant socioeconomic bias was detected.** Four fairness tests found no evidence that the model produced systematically larger prediction errors in disadvantaged neighborhoods. This result is limited to neighborhood-level predictions based on reported crime and therefore does not establish fairness for arrest-based or individual-level predictive policing.

These findings have several implications for policy. Crime data can help identify areas requiring attention, but raw crime rankings should not be treated as direct measures of neighborhood safety. Interventions should consider the wider socioeconomic conditions associated with crime rather than

focusing on individual indicators alone. Where predictive tools are used, human oversight, transparency, and regular independent fairness audits are necessary. Arrest-based data should also be treated cautiously because historical policing patterns can influence recorded arrests and create feedback loops in subsequent predictions.

The study has several limitations. Official crime data capture reported incidents rather than all victimization, and lower reporting rates in disadvantaged neighborhoods may mean that the recorded differences underestimate the true gap in crime exposure. The socioeconomic measures are based on ACS 5-year estimates and are treated as constant across the 60-month crime panel, so they cannot capture changes in neighborhood conditions over time. In addition, all variables are measured at the community-area level; therefore, the findings describe differences between areas and should not be interpreted as evidence about individual behavior, as this would risk the ecological fallacy. Finally, the fairness audit examines only calibration-style fairness for a model trained on reported crime. Its results therefore do not extend to other fairness measures, arrest-based models, or individual-level predictive policing tools.

Overall, the analysis provides evidence that crime in Chicago is shaped by both spatial and socioeconomic factors, while also demonstrating the importance of carefully evaluating how crime data are collected, modeled, and used. Predictive tools should therefore support, rather than replace, human judgment in decisions affecting communities.

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CRedit Author Statement

Endri Shenplaku: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

Generative AI (Claude, Anthropic) was only used as a coding aid in the implementation of data cleaning, visualization, clustering, predictive modeling procedures and debugging in Python. All code was reviewed and validated by the author, who can explain and defend it in full. GenAI was not used to generate analytical arguments, results interpretation, policy recommendations, or any other written content, all prose and conclusions are the author's own. The use of GenAI is disclosed in accordance with the course assignment guidelines.

The link to implemented code:

<https://github.com/EndriShen/dssc-crime-safety-chicago.git>